

Ultimate Nutrition

Amino Softgels

It has long been established that protein is an essential nutrient that needs to be obtained from our diets. Protein is an indispensable requirement for the growth and maintenance of any living creature. However, the ultimate value of a source of protein is its amino acid composition. A protein molecule is a long chain of amino acids linked by peptide bonds (i.e., an amino acid linked to another amino acid). Dietary protein is digested and absorbed into amino acids. Within the body, these amino acids play central roles both as building blocks of the body's proteins and as intermediates in metabolism, controlling virtually all cellular processes and reactions in living cells. Amino acids are responsible for the production of all the body's enzymes (including digestive enzymes), but they also play a key role in normalizing moods, concentration, aggression, attention, and sleep. Amino acids contribute significantly to the health of the nervous system, muscular structure, hormone production, vital organs, and cellular structure. More important, many physiological processes relating to exercise require amino acids for energy, recovery, muscle hypertrophy, and strength gains. Scientists, experts, and medical professionals all agree that getting enough amino acids in one's diet is an important factor in maintaining proper nutrition.

Twenty-two amino acids have been identified in humans that are naturally incorporated into the body's proteins. Of the twenty-two amino acids, eight are called "essential" amino acids (EAAs) because the human body cannot synthesize them from other compounds, so they must be obtained from our diet. The remaining fourteen amino acids are deemed "non-essential" because they can be made in the body. The branched chain amino acids (BCAAs), leucine, isoleucine, and valine, are classified among the EAAs and are very important for active individuals, bodybuilders, and professional athletes since they influence various aspects of muscle metabolism. BCAAs are of special importance to any nutritional program because they are not broken down in the liver but rather bypass it and are transported directly to the muscle.¹ BCAAs are unique in that they can be used either to build new proteins or be burned as fuel to produce energy.² Failure to obtain enough of even one of the essential amino acids results in degradation of the body's proteins.

It is important to note that the proteins that we consume in our diet are not equal. Protein-containing foods are grouped as either complete or incomplete proteins. A complete protein source (e.g., beef, chicken, fish, egg, milk, whey, and casein) is one that provides all of the essential amino acids, whereas an incomplete protein (e.g., rice, beans, grains, and legumes) lacks one or more of the essential amino acids. Plant proteins are typically combined to provide all of the essential amino acids and form a complete protein.³ As mentioned, the ultimate value of a food protein or a protein supplement is its amino acid composition. Amino Softgels from Ultimate Nutrition® constitutes a complete protein source since it provides all the essential amino acids. In terms of nutrition, it contains the right concentrations of the amino acids that the body cannot synthesize from other amino acids.

The human body requires a constant supply of protein and, thus, amino acids, as it is in a steady state of flux or turnover with both protein degradation (break down) and protein synthesis (build up). This state is referred to as protein balance. Protein balance is strongly dependent on the individual and lifestyle since protein is lost in urine, feces, blood, sweat, skin, nails, hair, etc. The amount of protein required for normal health is variable depending on many factors, such as body weight, age, physical activity, health condition, environment,

etc. Normally, the two processes are in equilibrium, with no net protein loss or gain. If protein synthesis is greater than protein degradation, the overall result is anabolic with a net increase in body protein called positive nitrogen balance. If protein degradation is greater than protein synthesis, the overall result is catabolic with a net decrease in body protein called negative nitrogen balance.

The largest amount of body protein is in the form of muscle. Any imbalance between the rate of synthesis and breakdown will lead to a change in the size of the muscle. When amino acid requirements are not met, muscle is broken down into amino acids, which are then sent to the amino acid pool to be used accordingly. When more protein is broken down than is synthesized, the body is in a negative nitrogen balance, and protein is lost. Therefore, it is important to ensure that you obtain the proper amount of protein throughout the day.

Resistance exercise is known to stimulate heightened changes in the rate of muscle protein turnover, resulting in an increase in both protein synthesis and degradation.⁴ Exercise causes a shift in this system because you're actually breaking down muscle tissue during this period, resulting in a negative nitrogen balance. During exercise, muscle protein is degraded along with increased oxidation (burning) of the BCAAs, signalling the body to stop protein synthesis in the muscle.⁵ When high concentrations of amino acids are present in the bloodstream during exercise, the anabolic response brought about by exercise is greatly increased. Therefore, there is an advantage to consuming amino acids pre-workout. After exercise, the body decreases its rate of protein synthesis and increases its rate of protein breakdown (proteolysis).⁶ Nitrogen balance becomes negative when more protein/amino acids are needed to help rebuild the structural aspects of muscle and recover from the stress of exercise, and in the absence of nutrition, the body will remain in negative nitrogen balance. Research has shown that consuming protein/amino acids before, during or after exercise can reverse this trend by increasing protein synthesis and decreasing protein breakdown.⁷⁻¹⁵ Failure to properly support exercise-induced recovery with the appropriate nutrient profile actually leads to muscle degeneration and performance decreases in the gym.

When most people hear the word protein or amino acid supplements, they usually assume protein powders. However, there are many different dietary protein supplements other than powders. One of the most convenient and often overlooked supplement is protein pills. The convenience of protein/amino acid pills can go a long way to support your protein requirements throughout the day and the times surrounding exercise. Amino Softgels by Ultimate Nutrition® is a high quality supplement that provides the EAAs and BCAAs. It can be taken anytime during the day to ensure that you meet your protein requirements.

FAQs

Q: What is the amount of amino acids found in Amino Softgels?

A: Amino Softgels provides 3000 milligrams of high quality amino acids including 500 milligrams of BCAAs per three-softgel serving.

Q: What amount of each amino acid is found in Amino Softgels?

A: Each three-softgel serving contains:

Essential Amino Acids	Amount	Non-Essential Amino Acids	Amount
L-Leucine*	236 mg	L-Arginine	116 mg
L-Isoleucine*	125 mg	L-Aspartic Acid	107 mg

L-Valine*	139 mg	L-Cystine	63 mg
L-Lysine	50 mg	L-Alanine	87 mg
L-Threonine	85 mg	L-Glutamic Acid	881 mg
L-Methionine	51 mg	L-Glycine	111 mg
L-Phenylalanine	183 mg	L-Histidine	68 mg
L-Tryptophan	27 mg	L-Proline	394 mg
		L-Serine	165 mg
		L-Tyrosine	112 mg
*Total Branch Chain Amino Acids		500 mg	
Total Essential Amino Acids (EAAs)		896 mg	
Total Amino Acids		3000 mg	

Q: Where have the amino acids in Amino Softgels been derived from?

A: The amino acids found in Amino Softgels have been derived from wheat and bovine gelatin.

Q: How do I take Amino Softgels?

A: Take 3 Softgels twice daily. For maximum results, take one serving pre-workout and one serving post-workout.

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